

0	7	.	1
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Describe how you would use cell fractionation techniques to obtain a sample of chloroplasts from leaf tissue. Do **not** include in your answer information about any solutions.

[3 marks]

[illegible]

0	7	.	2
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Table 3 shows features of a mitochondrion and a chloroplast. Complete **Table 3** with ticks where a feature is present.

[3 marks]

Table 3

Feature	Mitochondrion	Chloroplast
Double outer membrane		
Starch grains		
Diffusion of oxygen into the organelle		



0 7 . 3

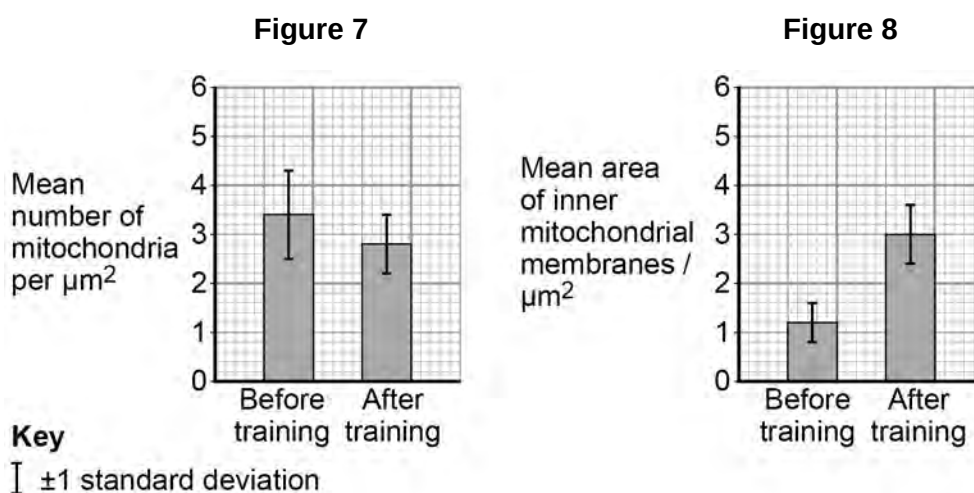
Give the function of a mitochondrion.

[1 mark]

0 7 . 4

Scientists investigated the effect of an exercise programme on the number and size of mitochondria in skeletal muscle. They took samples of muscle from a large number of volunteers before and after the exercise programme. From each sample, they cut thin sections and used these to determine the mean number of mitochondria per μm^2 and the mean area of inner mitochondrial membranes.

Their results are shown in **Figure 7** and **Figure 8**.



What do the data in **Figure 7** and **Figure 8** suggest about the effect of the exercise programme on mitochondria?

[2 marks]

